



6 BELLE and High-Speed Chess Circuitry

After completing his work on the UNIX operating system, an incredible accomplishment in itself and sufficient on which to rest one's laurels, Ken Thompson turned to developing a chess program. Thompson has a tremendous amount of energy, and when he becomes interested in something, exciting things happen. In addition to his work on computers, Thompson had other strong interests, including traveling to the South Pole, restoring antique autos, and, well, flying, about which a story cannot go unrecorded.

Thompson learned to fly in the late 1970s and flew one of his early solo flights to Montreal in a small two-passenger, single-winged plane to attend a conference on computer chess at McGill University. David Slate came too from Chicago, and when Thompson found out that both Slate and he were going on from Montreal to Carnegie Mellon University, Thompson invited Slate to fly there with him. Well, Slate has always been a very cautious person, someone who feels he is risking his life every time he boards a commercial airliner, and here Thompson wants him to go up in the sky on a plane that looks like it fought in the Battle of Normandy. Slate initially declined, but Thompson's charm eventually wore him down, and Slate finally accepted, much against his better judgment. Thompson was delighted with Slate's change of mind. Tony Marsland, who also attended the conference, drove with Thompson and Slate and me to the airport to send the fliers on



their way. Thompson's small plane was wheeled nearby to have gas put in and then pulled over to where we were waiting. The flyers got in; Thompson turned the key to start the engine, but the battery was dead as a door-nail. Slate was overjoyed. Maybe the mission was off! But no such luck. Thompson simply got out, cranked the propeller a few times, and the engine caught. Slate swallowed hard! We waved as they flew off, imagining Slate's knees knocking in the rear seat. This is not the end of the story. When Thompson finally arrived in Pittsburgh, the airport landing was very difficult. It was one of Thompson's first solo landings, a fact Slate probably didn't know. Thompson did his best, but somehow the plane wound up next to the runway on the grass but with everybody safe and sound. Rumor has it that Slate has stuck to commercial airlines ever since.

A decade after Thompson started his work on BELLE, his protégé became the first program crowned a master by the U.S. Chess Federation. He also was awarded the \$5,000 Fredkin Prize for having BELLE achieve this level of play before any other computer. This prize was established by Ed Fredkin's Fredkin Foundation of Cambridge, Massachusetts, in 1980. Fredkin also established a prize of \$100,000 to be awarded to the developers of the first program to defeat the human world champion in a regulation match. A \$10,000 prize was established for the developers of the first grandmaster-level computer, and as will be seen, this prize was picked up in 1989 by DEEP THOUGHT's team. Thompson began work on BELLE in 1973 and quickly came to the conclusion that his program could move to the head of the field if it could search much larger trees than the competition. At that time BELLE was searching about two hundred positions per second.

BELLE and special-purpose circuitry

Having the facilities of Bell Laboratories at his disposal, Thompson turned his energy toward developing special-purpose, high-speed circuitry for chess. Move generation is the most time-consuming process in chess programs and it was primarily this part of the code that he transformed into hardware. Working with design specialist Joe Condon, Thompson built three circuits: the first in 1977 with twenty-five chips, which gave virtually no speedup over the software version; the second a year or so later with 325 chips, which searched 5,000 positions per second; and a third shortly thereafter with 1,700 chips, which searched 120,000 positions per second.